

Discussion Module:

DSE Search, Debugging,
Sounds Like, Char Filters,
More

Use tools to validate/debug what is happening inside DSE Search / Apache Solr-

- Solr Admin screen
- Lucene index (Key) Explorer (LuKE)
- dsetool
- (CQLSH, Other)

Using these text analytics-

- Sounds Like
- Char Filters
- docValues (just part 1)

End to End Example:

Synonym, case insensitive

- DSE cluster, with Search enabled
- Authority to make tables, other
- CQLSH/Studio access-
- A modified version from the example in Discussion Module 7443.
- Added to, and we'll (validate/debug).



Example to test with .. page 1 of 12



```
DROP KEYSPACE ks_7445;
CREATE KEYSPACE ks_7445
  WITH REPLICATION =
    {'class': 'SimpleStrategy', 'replication_factor': 1};
USE ks_7445;
CREATE TABLE t1
(
  col1      TEXT,
  col2      TEXT,
  col3      TEXT,
  col4      TEXT,
  col5      TEXT,
  col6      TEXT,
  col7      TEXT,
  col8      TEXT,
  col9      TEXT,
  col0      TEXT,
  PRIMARY KEY ((col1, col2)) );
```

```
INSERT INTO t1
  (col1, col2, col3, col4, col5,
   col6, col7, col8, col9, col0)
VALUES ('aaa', 'aaa', 'Davie' , 'aaa', 'aaa',
        'aaa', 'aaa', 'aaa', 'aaa', 'aaa');

VALUES ('bbb', 'bbb', 'Elizabeth', 'bbb', 'bbb',
        'bbb', 'bbb', 'bbb', 'bbb', 'bbb');

CREATE SEARCH INDEX ON t1
WITH COLUMNS col1,
* { excluded : true };
```



Example to test with .. page 2 of 12

OLD
OLD
OLD

```
dsetool write_resource ks_7445.t1 name=7445_my_synonyms.txt file=7445_my_synonyms.txt
```

```
#  
# My synonyms file !!  
#  
Robert, Bob, Bobbie  
Dave, David, Davie, Dava Roonie  
Charles, Chuck, Charlie, Barkley  
Elizabeth, Beth, Liz, Lizzie, Tina
```



Prerequisites ?

Example to test with .. page 3 of 12

```
ALTER SEARCH INDEX SCHEMA ON t1
ADD types.fieldType[@name='TextField_17',
@class='solr.TextField',
@termVectors='false',
@termPositions='false',
@termOffsets='false',
@omitNorms='true'] with
{"analyzer":{"tokenizer":
{"class":"solr.StandardTokenizerFactory"},
"filter":{"class":"solr.SynonymFilterFactory",
"synonyms":"7445_my_synonyms.txt"}}};
```

```
ALTER SEARCH INDEX SCHEMA ON t1
ADD fields.field[@name='col3',
@type='TextField_17',
@indexed='true',
@multiValued='false',
@stored='true'];
```

col3



**OLD
OLD
OLD**

Example to test with .. page 4 of 12



```
RELOAD SEARCH INDEX ON t1;  
REBUILD SEARCH INDEX ON t1;
```

```
SELECT col1, col2, col3 FROM t1  
WHERE solr_query = '{ "q" : "col3:Dave" }' ;
```

```
WHERE solr_query = '{ "q" :  
  "col3:(Dava Rooney)" }' ;
```

```
WHERE solr_query = '{ "q" : "col3:dave" }' ;
```

```
INSERT INTO t1  
  (col1, col2, col3, col4, col5,  
   col6, col7, col8, col9, col0)  
VALUES ('aaa', 'aaa', 'Davie' , 'aaa', 'aaa',  
  'aaa', 'aaa', 'aaa', 'aaa', 'aaa');  
INSERT INTO t1  
  (col1, col2, col3, col4, col5,  
   col6, col7, col8, col9, col0)  
VALUES ('bbb', 'bbb', 'Elizabeth', 'bbb', 'bbb',  
  'bbb', 'bbb', 'bbb', 'bbb', 'bbb');
```

Dave, David, Davie, Dava Roonie
Charles, Chuck, Charlie, Barkley
Elizabeth, Beth, Liz, Lizzie, Tina

col3

Example to test with .. page 5 of 12



```
INSERT INTO t1
(col1, col2, col3, col4, col5,
 col6, col7, col8, col9, col0)
VALUES ('ccc', 'ccc', 'Stuart', 'Pizza Hut', 'ccc',
 'ccc', 'ccc', 'ccc', 'ccc', 'ccc');

VALUES ('ddd', 'ddd', 'Mary', 'Starbucks', 'ddd',
 'ddd', 'ddd', 'ddd', 'ddd', 'ddd');

VALUES ('eee', 'eee', 'Eli', 'Kranburrees', 'eee',
 'eee', 'eee', 'eee', 'eee', 'eee');
```

col4

Example to test with .. page 6 of 12

```
ALTER SEARCH INDEX SCHEMA ON t1
  ADD types.fieldType[@name='TextField18',
    @class='solr.TextField'] with
    {"analyzer":{"tokenizer":
      {"class":"solr.StandardTokenizerFactory"},
      "filter":{"class":"solr.BeiderMorseFilterFactory",
        "nameType":"GENERIC", "ruleType":"APPROX",
        "concat":"true", "languageSet":"auto"}}}};
ALTER SEARCH INDEX SCHEMA ON t1
  ADD fields.field[@name='col4',
    @type='TextField18',
    @indexed='true',
    @multiValued='false',
    @stored='true'];
RELOAD SEARCH INDEX ON t1;
REBUILD SEARCH INDEX ON t1;
```

col4



Example to test with .. page 7 of 12



```
SELECT col1, col2, col3, col4 FROM t1
```

```
WHERE solr_query = '{ "q" : "col4:(Pizzah Hutt)" }' ;
```

```
WHERE solr_query = '{ "q" : "col4:(Pizzah utt)" }' ;
```

```
WHERE solr_query = '{ "q" : "col4:(zah hut)" }' ;
```

```
WHERE solr_query = '{ "q" : "col4:Starrbux" }' ;
```

```
WHERE solr_query = '{ "q" : "col4:stARrbUx" }' ;
```

```
WHERE solr_query = '{ "q" : "col4:cranberry" }' ;
```

```
INSERT INTO t1
```

```
(col1, col2, col3, col4, col5,  
col6, col7, col8, col9, col0)
```

```
VALUES ('ccc', 'ccc', 'Stuart', 'Pizza Hut', 'ccc',  
'ccc', 'ccc', 'ccc', 'ccc', 'ccc');
```

```
VALUES ('ddd', 'ddd', 'Mary', 'Starbucks', 'ddd',  
'ddd', 'ddd', 'ddd', 'ddd', 'ddd');
```

```
VALUES ('eee', 'eee', 'Eli', 'Kranburrees', 'eee',  
'eee', 'eee', 'eee', 'eee', 'eee');
```

Example to test with .. page 8 of 12



```
ALTER SEARCH INDEX SCHEMA ON t1
  ADD fieldType[@name='TextField19',
    @class='solr.TextField'] WITH '{ "analyzer" :
{ "charFilter" :
  { "class" : "solr.HTMLStripCharFilterFactory" },
"tokenizer" :
  { "class" : "solr.StandardTokenizerFactory" }, "filter" :
    { "class" : "solr.LowerCaseFilterFactory" }}}';
```

```
ALTER SEARCH INDEX SCHEMA ON t1
  ADD fields.field[@name='col5',
    @type='TextField19'];
```

```
RELOAD SEARCH INDEX ON t1;
REBUILD SEARCH INDEX ON t1;
```

col5

Example to test with .. page 9 of 12



INSERT INTO t1

```
(col1, col2, col3, col4, col5,  
 col6, col7, col8, col9, col0)  
VALUES ('fff', 'fff', 'Bart', 'fff', 'my link2',  
 'fff', 'fff', 'fff', 'fff', 'fff');
```

```
VALUES ('ggg', 'ggg', 'Peggy', 'ggg',  
 'my <a href=\\"www.foo.bar\\">link</a>',  
 'ggg', 'ggg', 'ggg', 'ggg', 'ggg');
```

```
VALUES ('hhh', 'hhh', 'Earl', 'hhh',  
 '<bold>my link</bold>',  
 'hhh', 'hhh', 'hhh', 'hhh', 'hhh');
```

```
SELECT col1, col5 FROM t1 WHERE  
 solr_query = '{ "q" : "col5:(my link)" }' ;
```



Which rows ?
What col5 column value ?

col5

Example to test with .. page 10 of 12



```
ALTER SEARCH INDEX SCHEMA ON t1
  SET types.fieldtype[@class='org.apache.solr.schema.StrField']
    @name='TextField20';
ALTER SEARCH INDEX SCHEMA ON t1
  SET types.fieldtype[@class='org.apache.solr.schema.StrField',
    @docValues='true']@name='TextField21';
//
ALTER SEARCH INDEX SCHEMA ON t1
  SET types.fieldtype[@class='org.apache.solr.schema.TextField']
    @name='TextField22';
ALTER SEARCH INDEX SCHEMA ON t1
  SET types.fieldtype[@class='org.apache.solr.schema.TextField',
    @docValues='true']@name='TextField23';
```



What tokenizers ?

Example to test with .. page 11 of 12



```
ALTER SEARCH INDEX SCHEMA ON t1
  ADD fields.field[@name='col6',
    @type='TextField20'];
ALTER SEARCH INDEX SCHEMA ON t1
  ADD fields.field[@name='col7',
    @type='TextField21', @docValues='true'];
ALTER SEARCH INDEX SCHEMA ON t1
  ADD fields.field[@name='col8',
    @type='TextField22'];
ALTER SEARCH INDEX SCHEMA ON t1
  ADD fields.field[@name='col9',
    @type='TextField23', @docValues='true'];
```

```
RELOAD SEARCH INDEX ON t1;
REBUILD SEARCH INDEX ON t1;
COMMIT SEARCH INDEX ON t1;
```

col6, 7, 8, 9



Example to test with .. page 12 of 12



```
SELECT * FROM t1 WHERE solr_query =  
  '{"q":"*:*", "sort":"col6 asc"}'; // these both will fail  
SELECT * FROM t1 WHERE solr_query =  
  '{"q":"*:*", "sort":"col6 desc"}';
```

```
SELECT * FROM t1 WHERE solr_query =  
  '{"q":"*:*", "sort":"col7 asc"}';
```

```
SELECT * FROM t1 WHERE solr_query =  
  '{"q":"*:*", "sort":"col8 asc"}'; // this will fail
```

```
SELECT * FROM t1 WHERE solr_query =  
  '{"q":"*:*", "sort":"col9 asc"}';
```



End to End Example:

Synonym, case insensitive,
Review

- Prepare to validate/debug-



Apache Solr Admin UI

On by default. Can be locked down-

<http://localhost:8983/solr/#/>

<http://localhost:8983/solr/#/ks.tab>

<http://localhost:8983/solr/#/ks.tab> /...

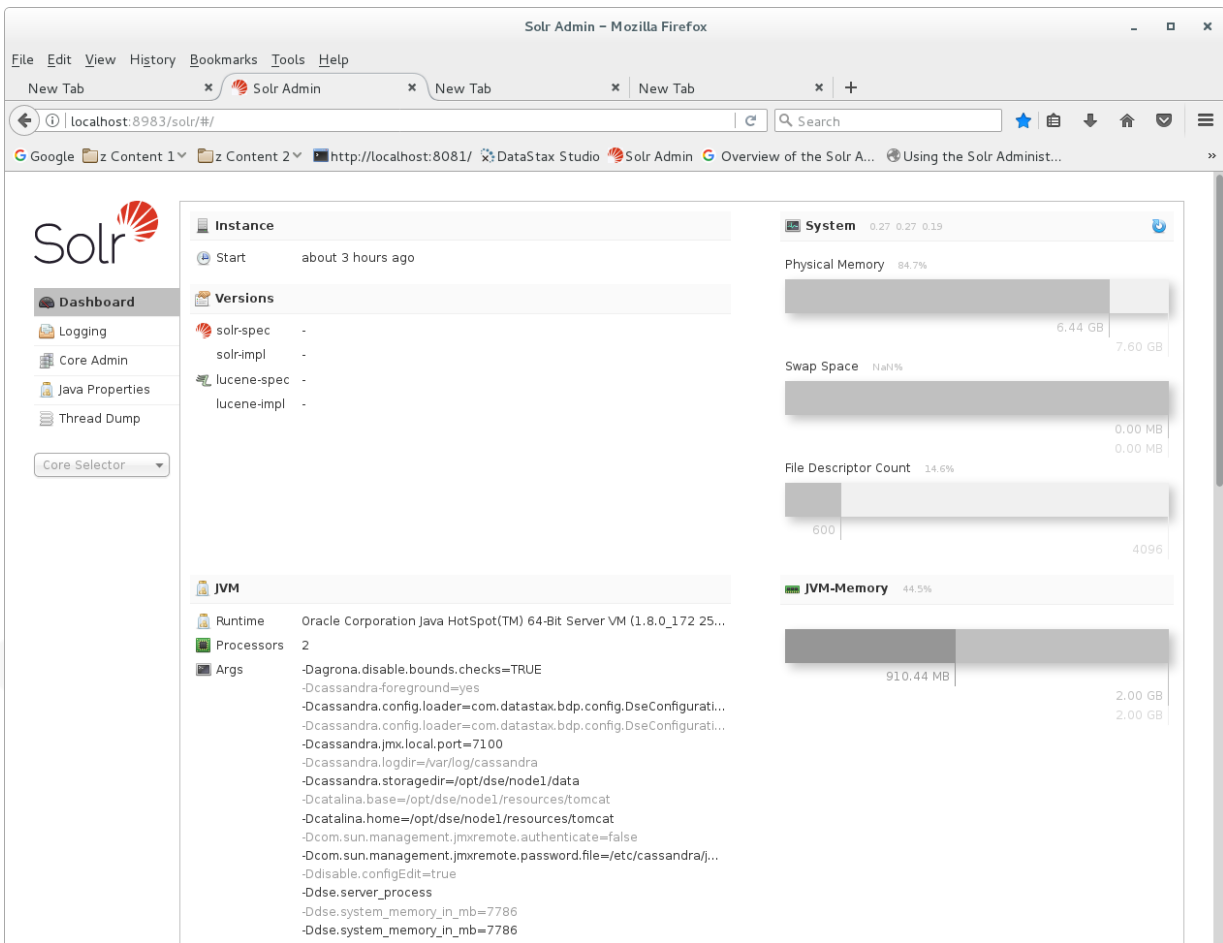
(Also REST API, Curl)

On line doc page,

<https://>

lucene.apache.org/solr/guide/6_6/overview-of-the-solr-admin-ui.html

- Read (write)
- Ad-hoc queries
- (System administration)
- Debugging/tracing
- Other



Solr Admin UI: Home page

- All 5 (first) TABs, read-only
- Dashboard; memory utilization
- (Remainder); Largely engineering debug

Solr Admin UI: After Core selection

Overview TAB-

- Row count
- Install base directory
- Data file directories

ks_7445.t1

- Overview
- Analysis
- Config
- Documents
- Files
- Ping
- Plugins / Stats
- Query
- Schema
- Schema Browser
- Segments info

Local Statistics

- Last Modified: 15 minutes ago
- Num Docs: 11
- Max Doc: 11
- Heap Memory: 3850
- Usage:
- Deleted Docs: 0
- Version: 19
- Segment Count: 1
- Current: ✔

Instance

- CWD: /opt/dse
- Instance: /opt/dse/solr
- Data: /opt/dse/node1/data/data/solr.data/ks_7445.t1
- Index: /opt/dse/node1/data/data/solr.data/ks_7445.t1/index
- Impl: org.apache.solr.core.StandardDirectoryFactory

Solr Admin UI: Core -> Analysis -> Column



Dashboard

Logging

Core Admin

Java Properties

Thread Dump

ks_7445.t1

Overview

Analysis

Config

Documents

Files

Ping

Plugins / Stats

Query

Schema

Schema Browser

Segments info

Field Value (Index)
Dave

Field Value (Query)

Analyse Fieldname / FieldType: col3 ?

☒ Verbose Output

[Analyse Values](#)

ST	text	Dave
	raw_bytes	[44 61 76 65]
	start	0
	end	4
	positionLength	1
	type	<ALPHANUM>
	position	1

SE	text	Dave	David	Davie	Dava	Roonie
	raw_bytes	[44 61 76 65]	[44 61 76 69 64]	[44 61 76 69 65]	[44 61 76 61]	[52 6f 6f 6e 69 65]
	start	0	0	0	0	0
	end	4	4	4	4	4
	positionLength	1	1	1	1	1
	type	<ALPHANUM>	SYNONYM	SYNONYM	SYNONYM	SYNONYM
	position	1	1	1	1	2

Documentation

Issue Tracker

IRI

- Debug existing analyzer
 - (Have to manually decompose: Technique)
- Index | Query | (Both)
- Displays (resource file data)

Solr Admin UI: Core -> Config

- *Shortcut to Files, Config*

- (Resource files)

- schema.xml

- DESCRIBE ACTIVE SEARCH INDEX SCHEMA ON ks.tab;



- solrconfig.xml

- DESCRIBE PENDING SEARCH INDEX CONFIG ON ks.tab;



http://localhost:8983/solr/ks_7445.t1/admin/file?file=7445_my_synonyms.txt&contentType=text/plain;charset=utf-8

7445_my_synonyms.txt
schema.xml
solrconfig.xml

http://localhost:8983/solr/ks_7445.t1

```
#  
# My synonyms file !!  
#  
Robert, Bob, Bobbie  
Dave, David, Davie, Dava Roonie  
Chuck, Charlie, Barkley  
h, Beth, Liz, Lizzie, Tina
```

```
<fieldType class="solr.TextField" name="TextField18">  
  <analyzer>  
    <filter class="solr.BeiderMorseFilterFactory" concat="true" languageSet="auto" nameType="GEI" />  
    <tokenizer class="solr.StandardTokenizerFactory" />  
  </analyzer>  
</fieldType>  
<fieldType class="org.apache.solr.schema.StrField" name="TextField20"/>  
<fieldType class="org.apache.solr.schema.StrField" docValues="true" name="TextField21"/>  
<fieldType class="org.apache.solr.schema.TextField" name="TextField22"/>  
<fieldType class="org.apache.solr.schema.TextField" docValues="true" name="TextField23"/>  
<fieldType class="solr.TextField" name="TextField24">  
  <analyzer>  
    <tokenizer class="solr.StandardTokenizerFactory" />  
  </analyzer>  
</fieldType>  
</types>  
<fields>  
  <field indexed="true" multiValued="false" name="col1" type="StrField"/>  
  <field indexed="true" multiValued="false" name="col2" type="StrField"/>  
  <field indexed="true" multiValued="false" name="col3" stored="true" type="TextField17"/>  
  <field indexed="true" multiValued="false" name="col4" stored="true" type="TextField18"/>  
  <field name="col5" type="TextField19"/>
```

Solr Admin UI: Core -> Plugins/Stats -> ..

core

class:	ks_7445.t1
version:	1.0
description:	SolrCore
src:	null

stats:

coreName:	ks_7445.t1
startTime:	2018-05-03T19:18:21.072Z
refCount:	2
instanceDir:	/opt/dse/solr
indexDir:	/opt/dse/node1/data/data/solr.data/ks_7445.t1/index/
indexSizeInBytes:	8317
indexSize:	8.12 KB
aliases:	ks_7445.t1

searcher

Searcher@2023496f[ks_7445.t1] main

- (Many, 8)
- Jars
- Resource per core, detailed

Solr Admin UI: Core -> Query

- \$\$\$
- Adhoc queries
- fl, *.score
- debugQuery (SET EXPLAIN)
- *Much more*

Request-Handler (qt)

/select

— common —

q

col4:starbuku

fq

sort

start, rows

0 10

fl

*, score

df

json.facet

Raw Query Parameters

key1=val1&key2=val2

wt

json

☒ indent

☒ debugQuery

☐ dismax

☐ edismax

http://localhost:8983/solr/ks_7445.t1/select?

```
{
  "responseHeader": {
    "status": 0,
    "QTime": 6
  },
  "response": {
    "numFound": 1,
    "start": 0,
    "maxScore": 4.2663274,
    "docs": [
      {
        "_uniqueKey": "[\"ddd\", \"ddd\"]",
        "col8": "ddd",
        "col9": "ddd",
        "col6": "ddd",
        "col7": "ddd",
        "col4": "Starbucks",
        "col5": "ddd",
        "col2": "ddd",
        "col3": "Mary",
        "col0": "ddd",
        "col1": "ddd",
        "score": 4.2663274
      }
    ]
  },
  "debug": {
    "rawquerystring": "col4:starbuku",
    "querystring": "col4:starbuku",
    "parsedquery": "SynonymQuery(Synonym(co",
    "parsedquery_toString": "Synonym(col4:s",
    "explain": {
      "[\"ddd\", \"ddd\"]": "\n4.2663274 = w",
    },
    "QParser": "LuceneQParser",
  }
}
```

Discussion Lab:

Discuss

Query Processing: ANDs and ORs-

The OR-topped query is a famous discussion point-

B-Tree+ Index: Query Processing

SELECT ... WHERE
col1 = x AND col2 = y;

SELECT ... WHERE
col1 = x OR col2 = y;



- **Assume B-Tree+**
- **Add any index**
- **How are rows processed**
 - (Table order)
 - Access methods
 - (Join methods) ?

Tf-IDF (Bitmap) Index: Query Processing

SELECT ... WHERE
col1 = x AND col2 = y;

SELECT ... WHERE
col1 = x OR col2 = y;

status = 'married'		region = 'central'		region = 'west'					
0		0		0		0		0	
1		1		0		1		1	
1	AND	0	OR	1	=	1	AND	1	=
0		0		1		0		1	
0		1		0		0		1	
1		1		0		1		1	

Image Source: https://docs.oracle.com/cd/E18283_01/server.112/e16579/indexes.htm



- Assume Tf-Idf (Bitmap)
- Add any index
- How are rows processed
 - (Table order)
 - Access methods
 - (Join methods) ?

End of Discussion Lab:

Solr Admin UI: Core -> Schema Browser

_uniqueKey

Field
_uniqueKey

Type
cql3CompositeKey_str

Unique Key Field
_uniqueKey

Field-Type: org.apache.solr.schema.StrField
Docs: 11

Flags:	Indexed	Tokenized	Stored	DocValues	Omit Norms	Omit Term Frequencies & Positions
Properties	✓			✓	✓	✓
Schema	✓			✓	✓	✓
Index	✓	✓	✓		✓	✓

? Index Analyzer: org.apache.solr.schema.FieldType\$DefaultAnalyzer

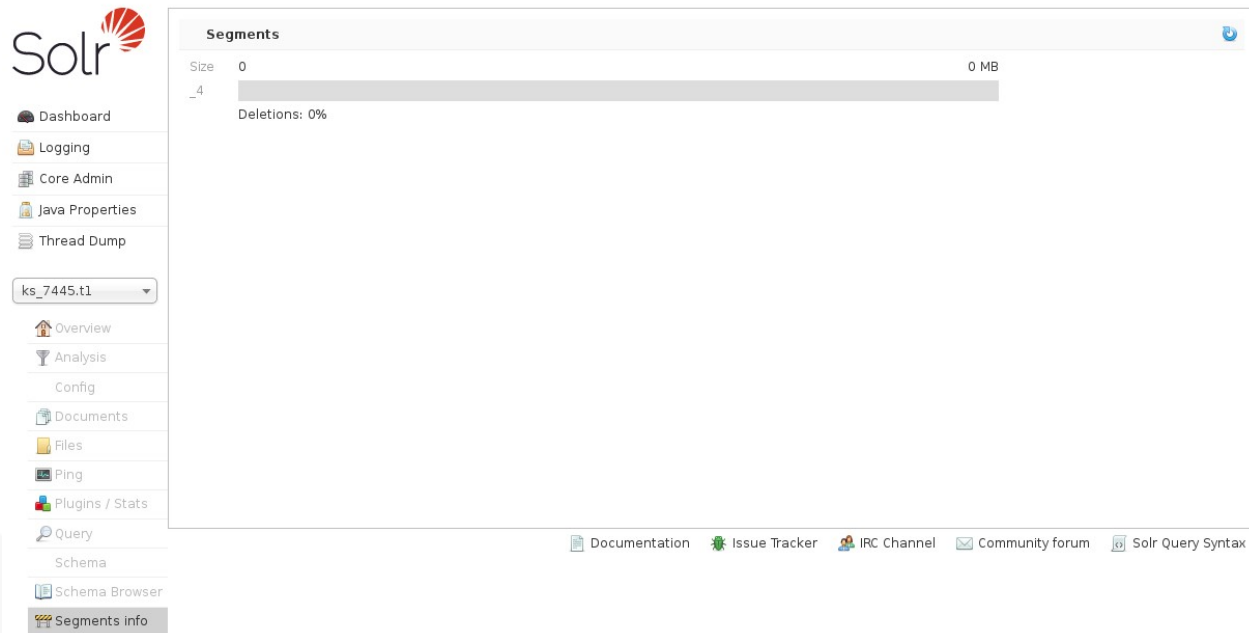
? Query Analyzer: org.apache.solr.schema.FieldType\$DefaultAnalyzer

i Load Term Info

- Similar to LuKE (later), some prefer LuKE (lower level)

Solr Admin UI: Core -> Segments Info

- SSTables segments, but for index data
- Important on volatile/large systems
 - (Compaction strategy)
 - (Deleting tombstones)



The screenshot displays the Solr Admin UI interface. On the left is a sidebar menu with the Solr logo at the top, followed by links to Dashboard, Logging, Core Admin, Java Properties, Thread Dump, Overview, Analysis, Config, Documents, Files, Ping, Plugins / Stats, Query, Schema, Schema Browser, and Segments info (which is highlighted). The main content area is titled 'Segments' and shows a table with one row for segment '_4'. The table columns are 'Size' (0) and 'Deletions: 0%'. A progress bar is visible below the table. At the bottom of the main area, there are links to Documentation, Issue Tracker, IRC Channel, Community forum, and Solr Query Syntax.

LuKE

Lucene index (Key) Explorer

- Recall that Lucene is the layer below Solr
- Should match DESCRIBE SEARCH INDEX
- Should match Solr Admin UI -> Core -> Schema Browser
- On line doc page,
<https://wiki.apache.org/solr/LukeRequestHandler>
- h ttp://localhost:8983/solr/ **ks.tab** /admin/luke
- h ttp://localhost:8983/solr/ **ks.tab** /admin/luke?id=aaa
- h ttp://localhost:8983/solr/ **ks.tab** /admin/luke?docid=1

LuKE: (display schema)

- docValues is particularly troublesome (many field types do not support it)
- fieldType -> field
 - Also properties related; Inheritance did it push forward (can it)

```
-<lst name="col8">
  <str name="type">TextField22</str>
  <str name="schema">IT-----</str>
  <str name="index">ITS-----</str>
  <int name="docs">11</int>
</lst>
-<lst name="col9">
  <str name="type">TextField23</str>
  <str name="schema">IT-D-----</str>
  <str name="index">ITS-----</str>
  <int name="docs">11</int>
</lst>
-<lst name="info">
  -<lst name="key">
    <str name="I">Indexed</str>
    <str name="T">Tokenized</str>
    <str name="S">Stored</str>
    <str name="D">DocValues</str>
    <str name="M">Multivalued</str>
    <str name="V">TermVector Stored</str>
    <str name="o">Store Offset With TermVector</str>
    <str name="p">Store Position With TermVector</str>
    <str name="y">Store Payloads With TermVector</str>
    <str name="O">Omit Norms</str>
    <str name="F">Omit Term Frequencies & Positions</str>
    <str name="P">Omit Positions</str>
    <str name="H">Store Offsets with Positions</str>
    <str name="L">Lazy</str>
    <str name="B">Binary</str>
    <str name="f">Sort Missing First</str>
    <str name="l">Sort Missing Last</str>
  </lst>
```

dsetool

- nodetool



- dsetool (--help)

- 180+ lines, many functions

- [core_indexing_status](#)

- create_core

- get_core_config

- get_core_schema

- [index_checks](#)

- infer_solr_schema

- [list_index_files](#)

- [perf](#)

- rebuild_indexes

- rebuild_core

- [stop_core_reindex](#)

- unload_core

- upgrade_index_files

- [write_resource](#)



**DESCRIBE
SEARCH INDEX**



**RELOAD |
REBUILD
SEARCH INDEX**

dsetool core_indexing_status ks.tab

```
# dsetool core_indexing_status ks_15.t1
[ks_15.t1]: INDEXING, 84% complete, ETA 18081 milliseconds (18
seconds)
# dsetool core_indexing_status ks_15.t1
[ks_15.t1]: INDEXING, 98% complete, ETA 2118 milliseconds (2
seconds)
# dsetool core_indexing_status ks_15.t1
[ks_15.t1]: FINISHED
```

Expect all lowercase identifiers for (Lucene/DSE Search) !

https://docs.datastax.com/en/dse/6.0/dse-dev/datastax_enterprise/tools/dsetool/dsetool.html

https://docs.datastax.com/en/dse/6.0/dse-dev/datastax_enterprise/tools/dsetool/dsetoolCore_indexing_status.html

dsetool index_checks ks.tab

dsetool index_checks ks_7445.t1

LUKE handler info:

numDocs:11
maxDoc:11
deletedDocs:0
indexHeapUsageBytes:3850
version:19
segmentCount:1
current:true
hasDeletions:false
directory:org.apache.lucene.store.MMapDirectory:
MMapDirectory@/opt/dse/node1/data/
data/solr.data/ks_7445.t1/index

...

...

lockFactory=org.apache.lucene.store.NativeFSLockFactory@2405262d
segmentsFile:segments_4
segmentsFileSizeInBytes:162
userData:{commitTimeMSec=1525441177156}
lastModified:Fri May 04 07:39:37 MDT 2018

dsetool list_index_files ks.tab

dsetool list_index_files ks_7445.t1

Filename	Encryption	Disk usage	Decrypted size	Encryption overhead
-----	-----	-----	-----	-----
_4.fnm	N/A	1.58K	N/A	N/A
_4.nvd	N/A	92 bytes	N/A	N/A
_4.nvm	N/A	118 bytes	N/A	N/A
_4.si	N/A	556 bytes	N/A	N/A
_4_BloomFilter_0.blm	N/A	119 bytes	N/A	N/A
_4_BloomFilter_0.doc	N/A	113 bytes	N/A	N/A
_4_BloomFilter_0.pos	N/A	80 bytes	N/A	N/A
_4_BloomFilter_0.tim	N/A	379 bytes	N/A	N/A
_4_BloomFilter_0.tip	N/A	105 bytes	N/A	N/A
_4_Lucene50_0.doc	N/A	124 bytes	N/A	N/A
_4_Lucene50_0.pos	N/A	223 bytes	N/A	N/A
_4_Lucene50_0.tim	N/A	3.19K	N/A	N/A
_4_Lucene50_0.tip	N/A	376 bytes	N/A	N/A
_4_Lucene54_0.dvd	N/A	550 bytes	N/A	N/A
_4_Lucene54_0.dvm	N/A	431 bytes	N/A	N/A
segments_4	N/A	162 bytes	N/A	N/A
write.lock	N/A	0 bytes	N/A	N/A

dse perf (options)

```
farrell0 — vi — 126x61

perf - Modify performance objects settings.
clustersummary      enable|disable  Toggle cluster summary statistics
cqlslowlog          enable|disable  Toggle CQL slow log threshold
                        Set the CQL slow log threshold,
                        recent_slowest_queries  Shows the N most recent slowest CQL queries
                        set_num_slowest_queries  Changes the number of slowest queries saved to be N
                        skip_writing_to_db  Keeps slow CQL queries in-memory only
                        write_to_db  Keeps slow CQL queries in-memory and also writes them to the DB

cqlsysteminfo      enable|disable  Toggle CQL system information statistics
dbsummary          enable|disable  Toggle database summary stats
histograms         enable|disable  Toggle column family histograms
resourcelatencytracking  enable|disable  Toggle resource latency tracking
solrcachestats     enable|disable  Toggle Solr cache stats
solrindexstats     enable|disable  Toggle Solr index stats
solrlatencysnapshots  enable|disable  Toggle Solr latency snapshots
solrrequesthandlerstats  enable|disable  Toggle Solr request handler stats
solrslowlog        enable|disable|threshold
                    Toggle Solr slow sub-query log or set its threshold once enabled

solrupdatehandlerstats  enable|disable  Toggle Solr update handler stats
userlatencytracking  enable|disable  Toggle user latency tracking
```

dsetool stop_core_reindex | write_resource ks.tab

- Graceful, minutes to wait-
- Not restartable

```
dsetool core_indexing_status ks_16.my_mapdata  
[ks_16.my_mapdata]: INDEXING, 90% complete,  
ETA 10950 milliseconds (10 seconds)
```

```
dsetool stop_core_reindex ks_16.my_mapdata  
Successfully stopped reindex for core  
ks_16.my_mapdata.
```

```
dsetool write_resource ks_7445.t1  
name=7445_my_synonyms.txt  
file=7445_my_synonyms.txt
```

Uploaded resource 7445_my_synonyms.txt for
ks_7445.t1

CQLSH/CQL

Describe, trace on|off

- DESCRIBE ACTIVE | PENDING SEARCH INDEX SCHEMA | CONFIG
- trace on | off (CQLSH only, not a CQL command)

CQLSH: tracing on | off

```
cqlsh:ks_7445> tracing off
```

Disabled Tracing.

```
cqlsh:ks_7445> tracing on
```

Now Tracing is enabled

```
cqlsh:ks_7445> select * from t1 where col3 LIKE 'aaa';
```

```
col1 | col2 | col0 | col3 | col4 | col5 | col6 | col7 | col8 | col9 | solr_query
-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----
(0 rows)
```

Tracing session: 1495de60-4fd3-11e8-98c6-630a381113b4

activity	timestamp	source	source_elapsed	client
Execute CQL3 query	2018-05-04 ...	127.0.0.1	0	127.0.0.1

Parsing select * from t1 where col3 LIKE 'aaa'; ...

Preparing statement ...

Processed response from shard 127.0.0.1:8609/solr/ks_7445.t1:

```
numFound: 0, maxScore: null, elapsed time: 5 ms,
token ranges: [(-9223372036854775808,-9223372036854775808)]
[Solr CQL query thread-0] | ...
```



End of Module:



Additional Detail:

Beider Morse Phonetic Matching



Image Source: http://horwitzfam.org/beider_and_book.htm
https://en.wikipedia.org/wiki/Alexander_Beider



Image Source: <https://upgs.wordpress.com/2007/11/02/steve-morse/>
https://en.wikipedia.org/wiki/Stephen_P._Morse